

Anomalous Transaction Detection for Financial Dark Pools

Kiara Jamal, Arya Kurup, Watson Blair

Abstract

This project focuses on exploring methods and strategies for detecting anomalous financial transactions. We combine statistical analysis with advanced clustering methods to create an anomaly detection pipeline that provides continuous, non-invasive, fraud monitoring.

Keywords

*Anomaly Detection, Clustering, Statistical Analysis,
Forensic Accounting,*

Introduction

Financial Crime is a serious and persistent international issue resulting in over \$3.1 trillion in illicit funds flowing through the financial system each year[1]. To combat this, persistent monitoring of financial transactions for anonymous or suspicious activity is a critical part of international law enforcement.

However, due to the volume of transactions, and a growing desire for instantaneous transaction processing; active examination of every transaction is not feasible. As such anomaly detection in this context is best achieved in two phases. First, large batches of transactions are checked for deviance from natural statistical distributions. Once a batch is identified as being likely to contain anomalous transactions, further examination with more time-intensive methodologies can be conducted.

(Call out that our pipeline is one piece of a broader fraud detection toolchain)

It specifically provides passive monitoring of transaction flows to identify suspect transactions, and needs to be used in conjunction with other tools including, rule-based AML engines, sanctions and watchlist screening, graph-based network analysis, case-management platforms, and KYC/Customer Due Diligence tools. These downstream systems provide the automated assessment required to determine whether flagged transactions represent genuine financial crime.

Related Concepts

The following topics are related to our exploration. We will provide a brief overview of the core topics, for more in-depth discussions please see our cited sources.

Benford's Law

Otherwise known as the first-digit law, Benford's Law is an observation that in many real-life sets of numerical data, the leading digit is likely to be small. In sets that obey the law, the number 1 appears as the leading significant digit about 30% of the time, while 9 appears as the leading significant digit less than 5% of the time.[2]

It is worth noting that in the United States, evidence based on Benford's law has been admitted in criminal cases at the federal, state, and local levels.[3] This cements it as a relevant indicator that deeper analysis of the dataset is warranted.

Financial Dark Pools

A financial institution dark pool is a private trading system, often operated by large broker-dealers or exchanges, that allows institutional investors to trade large blocks of securities anonymously without publicly displaying their orders. However, this foundation in anonymity and lack of transparency makes dark pool transactions inherently difficult to police and regulate.[4]

Gaussian Mixture Model

Gaussian Mixture Modeling(GMM) is an Expectation-Maximization(EM) algorithm that creates clusters based on the statistical features of the cluster: measure of center, variance, and previous probability. The resultant Probability Density Function(PDF) of the cluster defines the probability a given point is a member of the cluster.[5]

Isolation Forest

Isolation Forest is a clustering algorithm that uses binary trees to detect anomalous data. It works on the assumption that because anomalies are few and different from other data, they can be isolated using few partitions, driving Anomaly Score by the number of binary halvings of the dataset it takes to isolate a given point.[6]

It is relevant to note that IF can be overpowered if anomalous data is numerous enough to synthetically add density to an otherwise vacuous region of the feature space. In a real world application cross validation with divergent tool sets is required.[7]

Progressive Thresholding

In our context, Progressive Thresholding is used as an Expectation Maximization strategy. In each iteration of the algorithm we reexamine the set of points identified as non-anomalous for statistical properties consistent with non-anomalous datasets, if the set is still found to be anomalous we decrease our thresholds to include more points in the anomalous set.

This cycle continues until the remaining set is found to be statistically consistent with valid data, or the size of the set is reduced below statistically relevant thresholds. (Benford's law does not apply to small sets).

Methodology

Dataset

The dataset provided is incredibly well balanced, see Fig 1, with a slight preference for transactions in the first two quarters of the year.

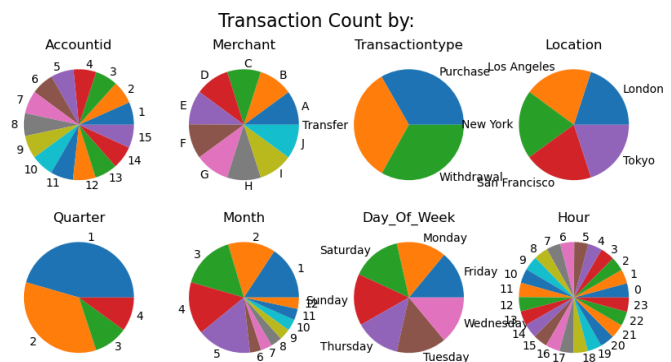


Fig. 1

The data consists of 216 thousand transactions executed during the 2023 calendar year. Transactions occur between 15 unique accounts and 10 unique merchants across 5 locations using 3 transaction types. All

merchants and accounts participate in all locations and transaction types. All accounts engage in transactions with all merchants. Transactions are representative of peer-to-peer interactions between a given Account and a given Merchant and can be of type 'Purchase', 'Transfer', or 'Withdrawal'.

We chose to conceptualize the dataset as inter-institutional transactions because the transaction types do not make sense if an individual is involved; an individual does not 'Withdrawal' from a retail merchant, nor do they traditionally make 'Purchase' from a financial institution.

However these transaction types do make sense if we cast the Merchants as securities exchanges and Account holders as brokerages where; 'Withdrawal's are seen as stock sales, 'Transfer's represent an 'In Kind' transfer of securities and 'Purchase's correlate to fulfilled purchase orders.

We further identified the transactions as Dark Pool Activity due to the anonymized nature of the transactions. We do not have information about what or how many of a given asset is being transacted upon; only the type of transaction and transactional total amount are visible.

Data Preparation and Processing

The data provided is complete and consistent, with no type-casting or data imputation required. This is in line with what we would expect for institutional transaction records. See Appendix 1 for a full description of the pre-processed dataset.

Feature Engineering and Encoding steps are facilitated by [CALL OUT FEATURE ENGINEERING and ENCODING UTILITIES] are supported by a set of custom utilities, including target encoding, label encoding, and one-hot encoding.

Anomaly Detection Pipeline

To provide continuous, non-intrusive, monitoring for anomalous activity we propose a Anomaly Detection Pipeline that uses low-cost bulk analysis to identify blocks of transactions which require further investigation.

Statistical Analysis is used to identify batches of transactions that are likely to contain fraudulent activity. Once given thresholds are surpassed, deeper investigation of the batch is triggered.

Indepth analysis consists of advanced clustering algorithms intended to identify the transactions most likely to be anomalous. Once anomalous data are removed from the batch, the remaining transactions are statistically re-examined to validate that anomalous data has been purged.

If statistical anomalies are still present, we repeat clustering tasks with increasingly stringent thresholds until the dataset presents as statistically consistent with non-fraudulent activity. The identified anomalous transactions are then flagged as suspicious and forwarded for further, specialized, analysis that is outside the scope of this project.

Statistical Analysis

The application of Benford's law to large batches of transactions allows us to 'smoke test' for fraud. Using Chi-Squared Distance, we measure the batch's divergence from statistical norms, allowing us to quickly survey transactions for anomalies.

If the distance is greater than our defined threshold of 15.507 at a 5% significance level(chance of false positive) further analysis is triggered to isolate anomalous records[8].

Clustering

Our pipeline facilitates anomaly detection cross referencing two clustering algorithms that examine data points from statistical and density based perspectives.

Gaussian Mixture Model(GMM) identifies outliers by measuring the probability of a query point's membership in a given cluster via a multi-dimensional probability density function. Points that do not satisfy a minimum membership threshold are identified as anomalous.

Isolation Forest(IF) identifies outliers by measuring the number of binary divisions it takes to isolate a given query point. Points that are isolated using fewer binary divisions are said to be anomalous due to their presence in a low-density area of the feature space. IF uses a contamination fraction term to specify the proportion of the data that is expected to be anomalous. This threshold specifies the percentage of the data that will be returned as anomalous.

We use the intersection of these two clustering algorithms to identify anomalous records.
[EXPOUND ON THIS]

Statistical Validation

After removing identified transactions from our batch, we re-evaluate the remaining set via Statistical Analysis. If the batch remains statistically anomalous we repeat cluster-based anomaly identification using progressively stringent thresholds.

This process continues until the remaining members are found to be statistically consistent with non-fraudulent data; or the population count is reduced below the threshold where statistical analysis is applicable. At which point the entire set is flagged for review.

If the batch is found to be statistically consistent with non-fraudulent transactions it is released for execution.

Results and Discussion

Our implemented pipeline can be viewed via the project's Github repository (<https://github.com/WatsonWBlair/cs667>).

Data was split into 52 batches, one per week, to simulate real-world monitoring conditions. Each batch progressed through our detection pipeline independently. Due to the statistical homogeneity of the data, we will use a selected batch as a representative of the whole population.

Evidence of Crime

As Fig. 2 clearly shows, our data is incredibly divergent from Benford's Law (depicted by the red line) with a Chi-Squared score of [SCORE HERE]. The divergence is significant enough to not only necessitate further analysis, but would most likely be sufficient indication of a financial crime to warrant a full investigation by regulatory authorities.

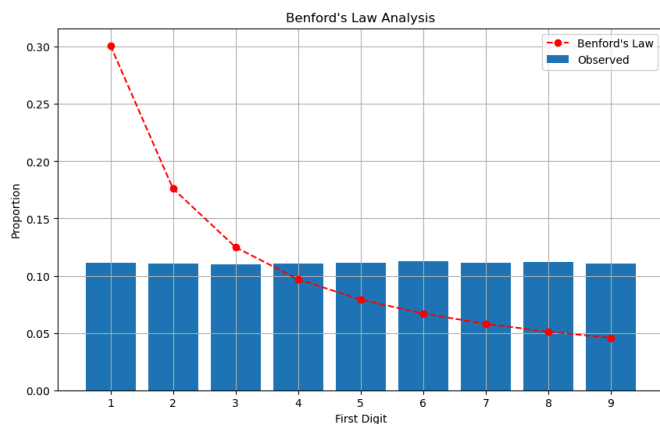


Fig. 2

This distribution of data is also indicative of the fully synthetic nature of our dataset.

Cross-Cluster Analysis

[NEED TO ACTUALLY RUN THE PIPELINE FOR RESULTS]

[DISCUSS HOW WE TAKE THE INTERSECTION OF IDENTIFIED POINTS TO BUILD SUBSET OF ANOMALOUS DATA]
[CAN WE USE GMM TO DRIVE THE CONTAMINATION FRACTION OF IF?]

Pipeline Results

Brief description of identified valid and anomalous data-sets.

If we get a 'non-anomalous' dataset out of the other side we should show the charts and graphs in an appendix.

Conclusion and Future Work

Continuing work would be to build more checks and clustering-cross validators into the pipeline. DBSCAN in particular can identify regions in the data that do not contain many transactions. (transaction regions that are low density)

What is our general view of the outcomes and findings?

What did we not have time to get to?

This is where we recommend that Sarbanes inject statistical features into the data. If tooling does not already exist for this purpose someone should make it...

The dataset seems to be a blank slate that would benefit from the injection of statistical features. Further efforts to refine the synthetic data and purposefully inject statistically relevant trends would greatly increase its utility as an instructional tool. Additionally modifying the transactional data to be more consistent with Benford's law increases the datasets fidelity to real-world scenarios and allows for the realistic fine-tuning of relevant thresholds.

Can we use Chi-Square score to drive starting thresholds?

Appendix

This section is reserved for discussion items that over-inflate the materials above.

We should include our statistical analysis of the dataset here - charts and graphs of [data.info\(\)](#) and `data.describe()` as well as charts showing the growth of anomaly detection over iterations, as well as batching details.

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